

Technology #117

Integrate a citrate delivery system into the venous filter that delivers anticoagulant before and after the filter chamber

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Integrate a citrate delivery system into the venous filter that delivers anticoagulant before and after the filter chamber

WHY DOES IT MATTER?

Documented recurring problem 'coagulation_in_device_or_device_ingredie' for venous filter: 17 MAUDE reports across 8 distinct devices (20140314 to 20230214); root-cause evidence from 0 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, custody-attached; the aggregation across records is AI_INFERENCE (bounded by the r Unresolved gap: Across the 2 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, device, devices, events, failure, function, intervention, introducing, maintain, malfunction, matching, modes, must, problem, producing, recurring,

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in ml_data. 7-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: In vitro testing comparing standard venous filters against modified filters with dual citrate delivery under simulated high-coagulation-r.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: held-out + temporal validation splits. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data; empirical risk on a non-representative sample says nothing about deployment risk (distribution shift); utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured; ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.; ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.; ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.; Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate [y_i , -] value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Dual-phase anticoagulation using citrate in both pre-filter and post-filter positions

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'coagulation_in_device_or_device_ingredie' for venous filter: 17 MAUDE reports across 8 distinct devices (20140314 to 20230214); root-cause evidence from 0 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, custody-attached; the aggregation across records is AI_INFERENCE (bounded by the r Unresolved gap: Across the 2 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, device, devices, events, failure, function, intervention, introducing, maintain, malfunction, matching, modes, must, problem, producing, recurring,

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: ml_data

Disciplines: data engineering, machine learning, software engineering, statistics

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: Integrate a citrate delivery system into the venous filter that delivers anticoagulant before and after the filter chamber

Subsystems:

- ?: data ingestion and labeling pipeline - ?
- ?: feature / representation layer - ?
- ?: model + uncertainty quantification - ?
- ?: deployment target + monitoring loop - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Integrate a citrate delivery system into the venous filter that delivers anticoagulant before and after the filter chamber

Key Physics: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: CONDITIONAL; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: NOT ESTABLISHED.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: CONDITIONAL; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: NOT ESTABLISHED.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: CONDITIONAL; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: NOT ESTABLISHED.

Assumptions:

- priors and likelihoods are MODEL_DERIVED until validated on deployment data
- empirical risk on a non-representative sample says nothing about deployment risk (distribution shift)
- utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured
- ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.
- ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.
- ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.
- Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.

- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate [y_i, -] value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'coagulation in device or device ingredie', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'distribution shift degrading accuracy post deployment', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'label noise propagating to biased predictions', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'overfitting to source-site artifacts', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'calibration drift invalidating decision thresholds', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'feedback-loop data contamination', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Devices matching 'venous filter' must maintain function under the conditions of the following malfunction events with	POD-001, Faulting
DI-002	Mechanism driver (proposed)	Dual-phase anticoagulation using citrate in both pre-filter and post-filter positions	MODELLED
DI-003	Expected effect target	Reduced coagulation within the venous filter while maintaining structural integrity and blood flow dynamics	MODELLED
DI-004	Domain design input pattern: decision applicability NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	invented applicability NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING PROFILE
DI-005	Domain design input pattern: availability applicability NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	invented applicability NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING PROFILE
DI-006	Domain design input pattern: latency applicability NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	invented applicability NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING PROFILE
DI-007	Regulatory design input: compliance candidate standards (FDA, CE, IEC, ISO, etc.)	compliance candidate standards (FDA, CE, IEC, ISO, etc.)	REGULATORY PROFILE
DI-008	Custodied observation: Perioperative deep vein thrombosis in elderly patients with lower extremity DVT	Perioperative deep vein thrombosis in elderly patients with lower extremity DVT	PERIOPERATIVE PROFILE
DI-009	Custodied observation: Recurrent lower extremity DVT in patients with lower extremity DVT	Recurrent lower extremity DVT in patients with lower extremity DVT	RECURRENT PROFILE
DI-010	Custodied observation: Comparison of anticoagulation strategies in patients with lower extremity DVT	Comparison of anticoagulation strategies in patients with lower extremity DVT	COMPARISON PROFILE
DI-011	Custodied observation: Efficacy and safety of anticoagulation strategies in patients with lower extremity DVT	Efficacy and safety of anticoagulation strategies in patients with lower extremity DVT	EFFICACY PROFILE
DI-012	Custodied observation: Case Report of severe pulmonary embolism in a patient with lower extremity DVT	Case Report of severe pulmonary embolism in a patient with lower extremity DVT	CASE REPORT PROFILE

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	data ingestion and labeling pipeline realizing Integrated data ingestion and labeling pipeline	CONCEPTUAL	integrated data ingestion and labeling pipeline
DO-002	feature / representation layer realizing Integrated feature / representation layer	CONCEPTUAL	integrated feature / representation layer
DO-003	model + uncertainty quantification realizing Integrated model + uncertainty quantification	CONCEPTUAL	integrated model + uncertainty quantification
DO-004	deployment target + monitoring loop realizing Integrated deployment target + monitoring loop	CONCEPTUAL	integrated deployment target + monitoring loop
DO-005	data ingestion and labeling pipeline interfaces with feature / representation layer	CONCEPTUAL	integrated data ingestion and labeling pipeline interfaces with feature / representation layer
DO-006	feature / representation layer interfaces with model + uncertainty quantification	CONCEPTUAL	integrated feature / representation layer interfaces with model + uncertainty quantification
DO-007	model + uncertainty quantification interfaces with deployment target + monitoring loop	CONCEPTUAL	integrated model + uncertainty quantification interfaces with deployment target + monitoring loop
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data processing	CONCEPTUAL	integrated device boundaries: patient/fluid inlet and outlet terminal control/data processing
DO-009	Flow/force/signal path geometry realizing the mechanism	CONCEPTUAL	integrated flow/force/signal path geometry realizing the mechanism
DO-010	Operating range for training data volume and provenance	UNKNOWN	which the mechanism holds its stated behavior for training data volume and provenance;
DO-011	Operating range for label quality rate within which the mechanism holds its stated behavior	UNKNOWN	which the mechanism holds its stated behavior for label quality rate; verification of range e
DO-012	Operating range for target operating point (precision/recall)	UNKNOWN	which the mechanism holds its stated behavior for target operating point (precision/recall);
DO-013	Operating range for calibration error budget within which the mechanism holds its stated behavior	UNKNOWN	which the mechanism holds its stated behavior for calibration error budget; verification of r
DO-014	Operating range for inference latency ceiling within which the mechanism holds its stated behavior	UNKNOWN	which the mechanism holds its stated behavior for inference latency ceiling; verification of
DO-015	Operating range for distribution-shift monitoring thresholds	UNKNOWN	which the mechanism holds its stated behavior for distribution-shift monitoring thresholds;
DO-016	No closed-loop control is proposed: the invention operates in open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	integrated no closed-loop control is proposed: the invention operates in open-loop as specified; this record exists so the omission is explicit and auditable
DO-017	Information pipeline: ingestion -> feature/representation -> decision	PROPOSED	integrated information pipeline: ingestion -> feature/representation -> decision
DO-018	Fixture enabling: held-out + temporal validation splits	UNKNOWN	which the mechanism holds its stated behavior for held-out + temporal validation splits; acceptance instrumentation
DO-019	Fixture enabling: calibration and decision-threshold analysis	UNKNOWN	which the mechanism holds its stated behavior for calibration and decision-threshold analysis; acceptance instrumentation
DO-020	Fixture enabling: failure-mode stress tests (distribution shift)	UNKNOWN	which the mechanism holds its stated behavior for failure-mode stress tests (distribution shift); acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: held-out + temporal split evaluation	UNKNOWN	Unknown; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
training data volume and distribution	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
label quality rate	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
target operating point (precision)	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
calibration error budget	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
inference latency ceiling	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
distribution-shift monitoring thresholds	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
coagulation in device	Failure by coagulation in device	single circuiting NOE E55 ABWS-100 (equipped with NOE E55 ABWS-100)	UNKNOWN; triggered when condition fails	data failure mode (Art.S
distribution shift degradation	Failing by distribution shift degradation	training and validation deployment of the model	UNKNOWN; triggered when condition fails	data failure mode (Art.S
label noise propagating	Failing by label noise propagating	target based prediction	UNKNOWN; triggered when condition fails	data failure mode (Art.S
overfitting to source	Failing by overfitting to source	subtle training data	UNKNOWN; triggered when condition fails	data failure mode (Art.S
calibration drift invalidation	Failing by calibration drift invalidation	loading decision threshold	UNKNOWN; triggered when condition fails	data failure mode (Art.S
feedback-loop data corruption	Failing by feedback-loop data corruption	external adjudication	UNKNOWN; triggered when condition fails	data failure mode (Art.S

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
coagulation in device or device degradation	Falsely by coagulation in device or device degradation	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)
distribution shift degrading accuracy	Falsely by distribution shift degrading accuracy	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)
label noise propagating to false predictions	Falsely by label noise propagating to false predictions	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)
overfitting to source-site artifacts	Falsely by overfitting to source-site artifacts	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)
calibration drift invalidating decisions	Falsely by calibration drift invalidating decisions	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)
feedback-loop data contamination	Falsely by feedback-loop data contamination	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	In vitro testing comparing standard ventus with test against standard ventus	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)	the device will not be used in the field (Article X)
VF-002	run killer experiment: falsification_test	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)	the device will not be used in the field (Article X)
VF-003	held-out + temporal validation splits. Acceptance criteria: (F1 score > 0.95)	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)	the device will not be used in the field (Article X)
VF-004	held-out + temporal validation splits. Acceptance criteria: (F1 score > 0.95)	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)	the device will not be used in the field (Article X)
VF-005	failure-mode stress tests (distribution shift)	design in the NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)	the device will not be used in the field (Article X)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	the device will not be used in the field (Article X)	the device will not be used in the field (Article X)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
NOT_APPLICABLE — data only invention: no physical materials exist to select; hardware dependencies are recorded under design	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
NOT	ESTABLISHED (no domain data)	NOT	ESTABLISHED (no domain data)				

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- MLOps CI for data and model: domain registry manufacturing route - candidate only; no process qualification exists
- versioned artifact registry with rollback: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter training data volume and provenance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter label quality rate [y_i -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter target operating point (precision/recall) [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter calibration error budget [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter inference latency ceiling [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter distribution-shift monitoring thresholds [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article implementation of the proposed physical verification process	per method	in vitro testing comparison	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	In vitro testing comparison	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	held-out + temporal variation	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	held-out + temporal variation	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	failure-mode stress test	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.